Po-Han (Kozak) HOU

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Programming: Python, C++, MATLAB, Julia, JavaScript
Scientific ML: Neural Operator (FNO, DeepONet), PINNs, PyTorch, TensorFlow, JAX
HPC & Systems: MPI, OpenMP, Slurm, PBS, Docker, Apptainer, CUDA
Domain Knowledge: Numerical Methods, Nuclear Fusion Plasma, Electromagnetic Waves, PDEs
Tools & Infra: HuggingFace, Cloudflare, ASR pipelines, LLM Deployment (Ollama)

PROFESSIONAL EXPERIENCE

Full Stack Engineer, AI Systems – XingAn Clinic — Yunlin, TW, 2025 ~ 2026
- Designed and deployed end-to-end clinic booking and queue management system integrated with LINE chatbot
- Built AI-assisted clinical pipeline (ASR + summarization) with privacy-aware, event-driven backend architecture

Software R&D Intern – Moldex3D (CoreTech) — Hsinchu, TW, 2023 ~ 2024
- Integrated Neural Operators (PINN, FNO) into production simulation API
- Achieved ~ 2000× speedup (hours → seconds) with <3% error
- Enabled real-time thermal simulation under varying boundary and material conditions

Deep Learning Research Intern – National Center for High Performance Computing — Hsinchu, TW, Jul 2022 ~ Aug 2022
(Supervisor: Dr. Chun-Yu Lin)
- Developed and benchmarked PINN, DeepONet, and FNO models on Xeon clusters and NVIDIA A100 DGX
- Implemented modular TensorFlow version of Fourier Neural Operator

EDUCATION

PhD in Nuclear Fusion Research Program — Chiba, JP (incoming, 2026)
University of Tokyo (Supervisor: Prof. Naoto Tsujii)
- Research Interests: Scientific ML, Fusion Plasma, Lower Hybrid Wave, Non-Inductive Current Drive, Digital Twins

MS in Applied Computational Science and Engineering — London, UK, 2025
Imperial College London (Supervisor: Prof. Adriana Paluszny and Prof. Naoto Tsujii)
- Research Interests: Modelling and Numerical Methods, Scientific Machine Learning (SciML)
- Dissertation: *Physics-based Machine Learning Scheme for Accelerating Simulation of 2D Lower-Hybrid Wave on TST-2 Spherical Tokamak (Merit)*

BS in Space Science and Engineering — Taoyuan, TW, 2024
National Central University (Supervisor: Prof. Jih-Hong Shue)
- Credit Program: Artificial Intelligence
- Cumulative GPA: 4.01 / 4.3 (Top 5%), Valedictorian

PUBLICATIONS

Hou, P.-H and Hsieh, S.-C. (2025) *Regression-based Physics Informed Neural Networks (Reg-PINNs) for Magnetopause Tracking Journal of Scientific Computing and Machine Learning, 1, 1-8. (Oral presentation, SCML 2025)*
- Achieved ~ 30% RMSE Reduction
- Cited by an IEEE Transaction paper and an IEEE conference paper

Hou, P.-H., Lin, W.-C., Hou, H.-C., Huang, Y.-H., Shue, J.-H. (2022*) Automatic Emergency Dust-Free solution on-board International Space Station with Bi-GRU (AED-ISS) arXiv:2210.08549.*
- Third Place, AI Space Challenge (ASEAN, 2022) – Best Technical Award

Chou, H.-H, Chiang, W.-J, Kuo, C.-W, Lung. C, **Hou, P.-H,** Wang, E.-J, Chen, Y.-T, Zhao, Y.-X, Kuo, J.-S (2021) *Utilizing Pop-Up Platform on the CubeSat to Achieve Commercial Activities in the Universe (UCCU) International Conference on Astronautics and Space Exploration (iCASE).*

HONORS & RECOGNITION

Third Place, NeurIPS 2024 ML4CFD Competition (Top 1% / 254 teams), 2024
- Achieved highest ML score (0.66) among all participants

Contributor, Google DeepMind Fusion Surrogates (PR#44), 2026
- QLKNN batch dimension alignment

Future Star Award (Best Student Poster), Space Science Society of ROC Annual Meeting, 2024
ICTP-IAEA College on Plasma Physics for Fusion Applications, ICTP, 2025
Level 2, Asia AI Odyssey Challenge (Microsoft Azure), 2024
Dean's Award (Valedictorian), College of Earth Science, National Central University, 2024
TensorFlow Developer Certificate, Google, 2023